

$$= [x+y \cdot z] \text{ (Theorem 6(a)) } \square$$

$$(b) xy + xz + yz' = xz + yz' \text{ (Postulate 2(b))}$$

Solution:

$$\begin{aligned}
 xy + xz + yz' &= (xy \cdot 1 + xz + yz') \text{ (Postulate 5(a))} \\
 &= (xy \cdot (z+z') + xz + yz') \text{ (Postulate 5(a))} \\
 &= (xyz + xy z' + xz + yz') \text{ (Postulate 4(a))} \\
 &= (xz + (xyz + xy z')) + yz' \text{ (Postulate 3(a))} \\
 &= (xz + xy z) + xy z' + yz' \text{ (Theorem 4(a))} \\
 &= (xz + xy z) + (xy z' + yz') \text{ (Theorem 4(a))} \\
 &= (xz + x \cdot (yz)) + (x \cdot (yz') + yz') \text{ (Theorem 4(b))} \\
 &= (xz + x \cdot (zy)) + (x \cdot (yz') \cdot x + yz') \text{ (Postulate 3(b))} \\
 &= (xz + x \cdot (zy)) + (yz') \cdot x + yz' \text{ (Theorem 4(b))} \\
 &= (xz + (x \cdot z) \cdot y) + (yz') \cdot x + yz' \text{ (Theorem 4(b))} \\
 &= xz + yz' \text{ (Theorem 6(a)) } \square \rightarrow \text{Must apply Postulate 3(a).}
 \end{aligned}$$

2.4 Boolean Functions

- A binary variable can take the value of 0 or 1.
- A boolean function is an expression formed with binary variables, the two binary operators OR and AND, the unary operator NOT, parentheses, and equal sign.
- For a given value of the variables, the function can be either 0 or 1.
- A Boolean function may also be represented in a truth table. To represent a function in a truth table, we need a list of the 2^n combinations of 1's and 0's of the n binary variables, and a column showing the combinations for which the function is equal to 1 or 0.

Table 2-2 Truth tables for $F_1 = xyz'$, $F_2 = x + y'z$, $F_3 = x'y'z + x'yz + xy'$, and $F_4 = xy' + x'z$

x	y	z	F_1	F_2	F_3	F_4
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	1	0	1
0	1	1	0	1	1	1
1	0	0	0	0	0	0
1	0	1	1	0	0	0
1	1	0	1	1	1	1
1	1	1	1	1	0	0